

Programming in Java - Solution Guide

This PDF contains concise solutions and explanations for the uploaded Java Programming question paper.

PART A – Short Answers

Question	Answer
Q1(a)	Inheritance is the technique of creating new classes using existing class behavior.
Q1(b)	TreeSet stores unique elements in sorted order.
Q1(c)	Comparable and Comparator APIs are used for sorting objects.
Q1(d)	A thread is a single execution path, while multithreading allows multiple threads to execute simultaneously.
Q1(e)	Exception is an abnormal condition during execution. Types: Checked and Unchecked exceptions.
Q1(f)	HSV uses Hue, Saturation, Value whereas HSL uses Hue, Saturation, Lightness.
Q1(g)	Layout Managers arrange GUI components automatically in containers.
Q1(h)	JDBC is used to connect Java applications with databases.
Q1(i)	Statement executes normal SQL queries, while PreparedStatement executes precompiled parameterized queries.
Q1(j)	URL parts: Protocol, Domain, Port, Path, Query Parameters.

PART B – Detailed Solutions

Q2

```
Correct Order:
public class Welcome {
    public static void main(String[] args) {
        System.out.println("morning");
        System.out.println("afternoon");
    }
}
```

Q3

Lambda Expressions provide concise syntax for functional programming in Java.

Types:

1. No parameter lambda
2. Single parameter lambda
3. Multiple parameter lambda

Difference:

Functions are reusable blocks, whereas lambda expressions are anonymous functions.

Q4

CRUD Operations in JDBC:

Create – INSERT
Read – SELECT
Update – UPDATE
Delete – DELETE

Example:

```
Connection con = DriverManager.getConnection(url,user,password);
Statement st = con.createStatement();
```

```
st.executeUpdate("INSERT INTO student VALUES(1,'Mohan')");
```

Q5

Formatted strings can be created using `String.format()`.

Difference:

`String` - Immutable

`StringBuffer` - Mutable and thread-safe

`StringBuilder` - Mutable and faster than `StringBuffer`

Q6

Thread communication methods:

`wait()` - pauses thread

`notify()` - wakes one thread

`notifyAll()` - wakes all waiting threads

Q7

`DatabaseMetaData` provides database information such as:

- Database name

- Driver version

- Table details

- Supported SQL features

Methods:

`getTables()`, `getColumns()`, `getDriverName()`, `getURL()`

Q8

`DatagramPacket` is used in UDP communication.

Constructors:

`DatagramPacket(byte[] buf, int length)`

Methods:

`getData()`

`getLength()`

`getAddress()`

Q9

Checked Exception:

Occurs at compile time.

Example: `IOException`

Unchecked Exception:

Occurs at runtime.

Example: `ArithmeticException`

Q10

`BoxLayout` arranges components horizontally or vertically.

Example:

```
setLayout(new BoxLayout(panel, BoxLayout.Y_AXIS));
```

Q11

`JSlider` is used to select values graphically.

Constructors:

`JSlider()`

`JSlider(int min, int max)`

Methods:
setValue()
getValue()
setMajorTickSpacing()